Pick and Place of a Cube with a Franka Panda
Mink vs DLS-IK vs QP-IK

Lab Tutorial -3

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1. Introduction and goals

A Franka Emika Panda in MuJoCo picks a 5 cm cube placed at a random position and yaw on a table and drops it into a tray. The pipeline was built first with an off-the-shelf IK library (mink), then with two custom IK methods i.e Damped Least Squares (DLS-IK) and Quadratic Programming with joint-limit constraints (QP-IK). All three were compared on the same 25 seeded episodes. The goals were to analyse the task and dexterous workspace and size the table from it, randomize and log episodes at scale, analyse failures and mitigate them and compare the three IK methods on success, planning time, IK iterations, clearances and constraint violations.

2. Method

2.1 Environment and table design

The scene is the MuJoCo Menagerie Panda (base at the origin), a table, a 5 cm, 80 g cube on a free joint and a fixed open tray (14 × 14 cm outer, 12 × 12 cm cavity, 8 cm walls) centred at (0.45, 0.15) m on the table. The Tool Center Point (TCP) is the centre of the fingertip, 0.103 m below the hand body.

Parameter	**Symbol**	**Value**	**Reason**
Table length (x)	L	0.60 m, (x 0.20–0.80)	Within dexterous workspace
Table width (y)	W	0.50 m, (y −0.25–0.25)	
Table height	H	0.40 m (top surface)	

2.2 Episodes and task phases

Variable	Range	Notes
x	[0.24, 0.76] m	Table surface minus a 4 cm edge band
y	[−0.21, 0.21] m	Same band; 30% of spawns drawn from a 6 cm ring around the tray
yaw	[−π, π]	Uniform

All methods run the same 25 episodes (seed 0). Each episode moves the TCP through pre-grasp (12 cm back along the approach axis), descend, close, lift, transit, lower, release and retreat. Targets are interpolated with a smoothstep profile and tracked every 10 ms by position actuators with gravity compensation. Each move ends with convergence gating: the target is held until the TCP is within 5 mm (at most 2 s). Grasp selection is same for all methods. A cube repeats every 90°, so the gripper yaw is the cube yaw plus k·90°, taking the k closest to the home yaw (least wrist rotation). If planning fails, the other three yaws are tried, then a tilted approach (15°, 30°, 45° about the finger-closing axis, hand on the robot side). A candidate is accepted only if every waypoint converges and keeps ≥ 5 mm robot clearance to the table and tray.

2.3 IK methods. All three use the same task error.

- **Mink (baseline):** a FrameTask on the hand plus a weak PostureTask, solved with DAQP. Five iterations per control step update mink's internal configuration, which is sent to the actuators (open-loop reference).

- **DLS-IK (closed-loop):** from the measured state every step, with the given secondary task. λ rises from 10^{-4} to 6.4×10^{-3} as manipulability drops below 0.02. Speeds are scaled to ≤ 1.5 rad/s and the command is clipped to the joint limits, because DLS has no constraints.

- **QP-IK (closed-loop):** the same task error as a QP, with joint-position limits as a velocity damper ($K_{lim} = 5\ s^{-1}$) and joint-speed limits (1.5 rad/s), in the tutorial's $G\,\dot{q} \le h$ form.

2.4 Collision avoidance

- **Robot vs table and tray.** every planned pose must keep ≥ 5 mm clearance, and any contact during execution ends the episode as collision.

- **Cube vs tray:** the transit height keeps the cube bottom 5 cm above the tray walls, and release happens over the cavity centre 1 cm above the walls. Cube–tray clearance is measured during lift, transit and lower, and any contact before release ends the episode as cube_tray_collision.
- **Distances:** computed as exact point-to-box signed distances from points sampled on the collision meshes (mean error 0.2 mm).

3. Results

On the same 25 episodes, Mink and DLS-IK each succeed 23 times (92%) and QP-IK 20 times (80%). No method hit a joint limit or exceeded a Panda joint-velocity limit.

Metric	Mink	DLS-IK	QP-IK
Success rate	92% (23/25)	92% (23/25)	80% (20/25)
Mean planning time per episode	1.04 s	0.30 s	0.26 s
IK iterations per converged waypoint	2.6	3.6	4.1
Planning IK iterations per episode (mean)	1144	1254	1621
IK compute per 10 ms control step	3.33 ms	0.25 ms	0.20 ms
Min robot–table / robot–tray clearance	13.9 / 4.8 mm	11.0 / 1.4 mm	0.9 / 6.1 mm
Min cube–tray clearance	21.4 mm	21.5 mm	21.6 mm
Max TCP error while moving / settled	24.7 / 3.4 mm	58.4 / 4.8 mm	157.0 / 5.0 mm
Failures	1 IK infeasible, 1 planned collision	1 IK infeasible, 1 planned collision	2 IK infeasible, 2 planned collision, 1 collision

- **Accuracy:** all three converge to ≤ 5 mm at the end of every move. The convergence gate adds 0.02–0.05 s per episode.

- **Computation:** the custom closed-loop methods need one 7 × 7 solve (DLS) or one small QP (QP) per step, 13–17× less compute than five mink iterations. Mink's planning is 3–4× slower for the same reason (Python overhead per iteration).

- **Speed:** the 1.5 rad/s cap keeps DLS and QP at 43% of the Panda's joint-speed limits; mink's open-loop reference reaches 72%. The cap is why the closed-loop methods lag on large reconfigurations (up to 58 mm for DLS and 157 mm for QP while moving), although they settle on target.

- **Cube–tray:** the transit and release heights keep the cube at least 21 mm from the tray walls in every episode, with no cube–tray contact before release.

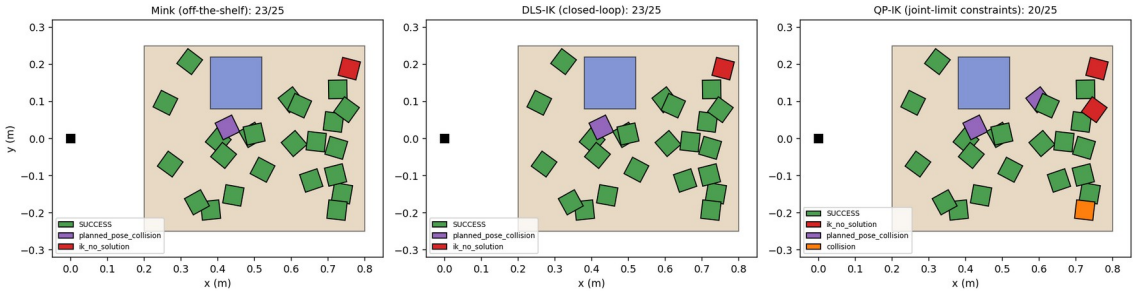


Figure 1. Every episode for each method, cubes drawn at their sampled position and yaw. All three fail at episodes 9 (cube 3 cm from the tray wall) and 17 (far corner). QP additionally fails at 6, 15 and 24.

4. Failure Analysis and Mitigation

Without mitigation, Mink fails 15 of the 25 episodes (40% success), all at planning. With the grasp mitigation the same planner fails 2 (92%).

5. Conclusions

The grasp strategy mattered more than the IK method: the mitigation moved Mink from 40% to 92% success, while the three IK methods differ by at most 3 episodes.

Method	Strengths observed	Weaknesses observed
Mink	Fewest iterations per waypoint (2.6), robust to wrist-limit cases, highest success with DLS	13–17× more compute per control step (3.3 ms); open-loop reference, so the TCP trails by up to 25 mm while moving
DLS-IK	Simple closed form, 0.25 ms per step, adaptive damping keeps it stable at the workspace boundary, 92% success	No constraints: at a joint limit it can only clip, so it stalls (it failed yaw 225° at the tray, where joint 7 needs 2.90 rad); clipping distorts the TCP path
QP-IK	Joint-position and speed limits enforced inside the optimisation; cheapest per step (0.20 ms); with a software joint limit it kept TCP error at 3.7 mm where DLS degraded to 24 mm	When speed limits saturate it gives up task accuracy (157 mm lag, ep 24 collision); more local minima in planning (2 IK failures vs 1)

- **Workspace and table:** a vertical gripper covers 72% of the 0.60 × 0.50 m table. The far strip needs a tilted approach, and the near edge must stay ≥ 0.20 m from the base to clear links 1–2.

- **Cube–tray avoidance:** a transit height 5 cm above the walls and release over the cavity centre kept the cube ≥ 21 mm from the tray in every episode.

- **Closed-loop vs open-loop:** closed-loop DLS and QP settle within 5 mm with an order of magnitude less computation, but they need convergence gating and slower phases for large reconfigurations.

- **Constraints:** QP-IK is the right tool when joint or speed limits are active. Its failure in ep 24 shows that a collision constraint, not only joint limits, belongs in the QP.